

Lecture Note

Time Series: Heteroskedasticity and Serial Correlation

- (A) Heteroskedasticity
- (B) Serial Correlation
- (C) Correcting for Serial Correlation

(A) Heteroskedasticity

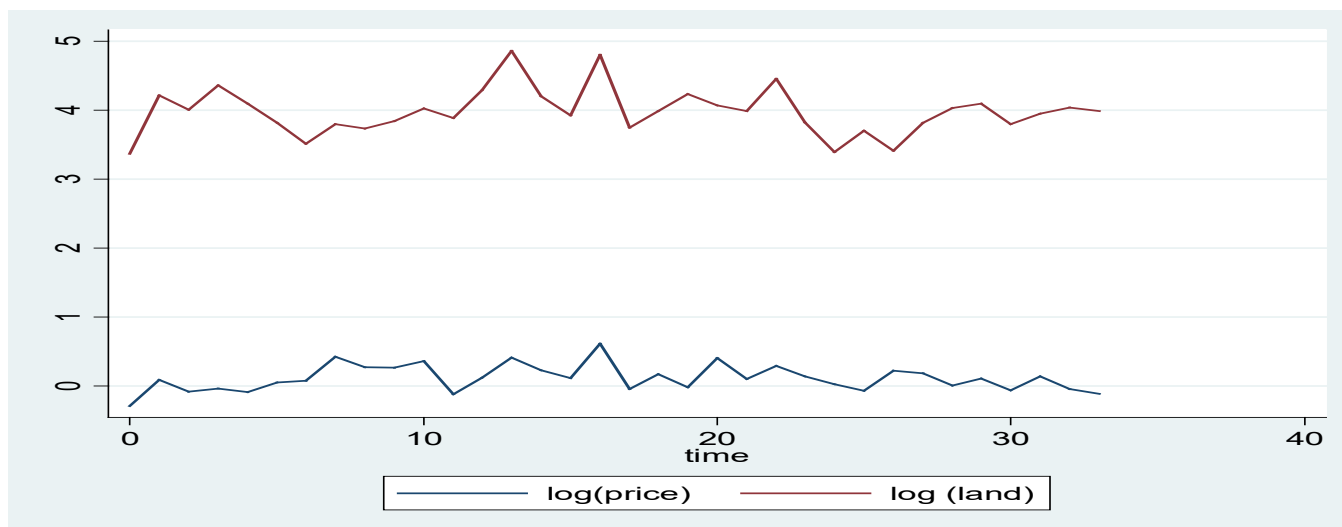
- We discussed testing and correcting for heteroskedasticity in the cross-sectional models (refer to Lecture Note 2). Heteroskedasticity can also occur in time series regression models. Heteroskedasticity does not cause bias or inconsistency in the $\hat{\beta}_j$. But it invalidates the usual standard errors, t statistics, and F statistics. The same conclusion holds for the time series models.
- Standard errors for the estimated regression coefficients are wrong (i.e., bias) and the direction of the bias is not known a priori. In addition, the wrong standard errors are often smaller.
- We can test for heteroskedasticity in time series regressions. But when we use the test learned from the previous course, we need to make sure first that the errors u_t is serially uncorrelated. Any serial correlation will generally invalidate a test for heteroskedasticity.

- In practice, one should always test for serial correlation. After something has been done to correct for serial correlation, one can then test for heteroskedasticity.

Empirical example:

When price of sugarcane is high, farmers plant more than when its price is low. We are therefore interested in the regression model: $\ln(Area_t) = \beta_0 + \beta_1 \ln(price_t) + u_t$

Information on β_1 is useful for government planning. Because, it is important to know whether existing sugar processing mills are likely to be able to handle predicted output, whether a pricing policy linking production, processing, and consumption is desirable.



. reg Lland Lprice

Source	SS	df	MS
Model	.741225795	1	.741225795
Residual	3.03157079	32	.094736587
Total	3.77279658	33	.114327169

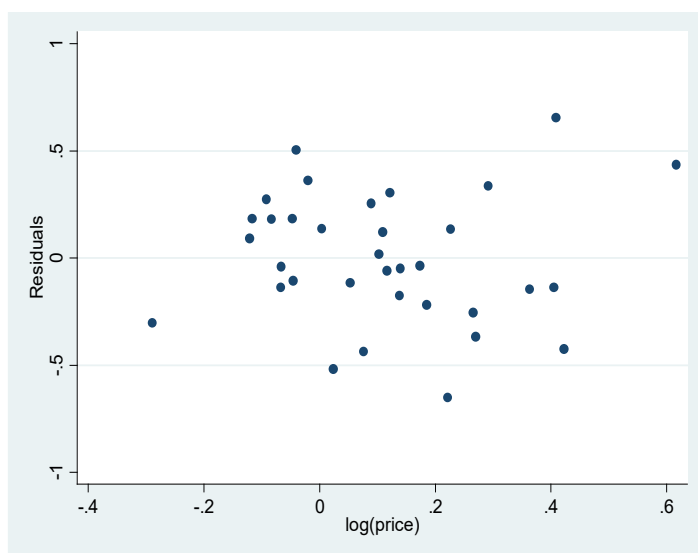
Number of obs = 34
 F(1, 32) = 7.82
 Prob > F = 0.0087
 R-squared = 0.1965
 Adj R-squared = 0.1714
 Root MSE = .30779

Lland	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
Lprice	.7761187	.2774673	2.80	0.009	.2109364	1.341301
_cons	3.893256	.0613451	63.46	0.000	3.7683	4.018212

- How do we know Assumption TS. 4: $\text{Var}(u_t|X) = \text{Var}(u) = \sigma^2$ holds?

> **predict** uhat, residual

> **graph twoway scatter** uhat Lprice



- It seems that variation of the residual, u does not depend on the values of x .
But we can use a more scientific way to examine $\text{Var}(u_t|X) = \text{Var}(u)$.

- **Testing Homoskedasticity:** $Var(u|x_1, \dots, x_k) = \sigma^2$

Consider a regression model:

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \dots + \beta_k x_{tk} + u_t, \quad t = 1, \dots, n$$

We would like to test $H_0: Var(u|x_1, \dots, x_k) = \sigma^2$ against $H_1: H_0 \text{ is wrong}$.

Recall: $Var(u|x_1, \dots, x_k) = E(u^2|x_1, \dots, x_k) - [E(u|x_1, \dots, x_k)]^2$

- Testing $H_0: Var(u|x_1, \dots, x_k) = \sigma^2$ is same as testing $E(u^2|x_1, \dots, x_k) = \sigma^2$
- Conceptually, the null hypothesis means that u^2 (in expected value sense) is not related to one or more of explanatory variables $\mathbf{x}$, because σ^2 is constant and not a function of $\mathbf{x}$.

- We can use the four-step procedure (refer to Lecture Note 2). In the fourth step, we run the following regression model.

$$H_0: \delta_1 = 0, \delta_2 = 0, \dots, \delta_k = 0 \text{ against } H_1: H_0 \text{ is wrong}$$

$$\hat{u}^2 = \delta_0 + \delta_1 x_1 + \dots + \delta_K x_K + v$$

. reg uhat2 Lprice

Source	SS	df	MS	Number of obs = 34		
Model	.040223922	1	.040223922	F(1, 32)	=	3.37
Residual	.382249102	32	.011945284	Prob > F	=	0.0758
Total	.422473024	33	.012802213	R-squared	=	0.0952
				Adj R-squared	=	0.0669
				Root MSE	=	.10929

uhat2	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
Lprice	.1807986	.0985261	1.84	0.076	-.0198924	.3814897
_cons	.0687983	.0217831	3.16	0.003	.0244276	.1131689

Step 1: Regress y on x

Step 2: Predict the residual using $\hat{u} = y - \hat{y}$

Step 3: Generate $\hat{u}$ by taking the square

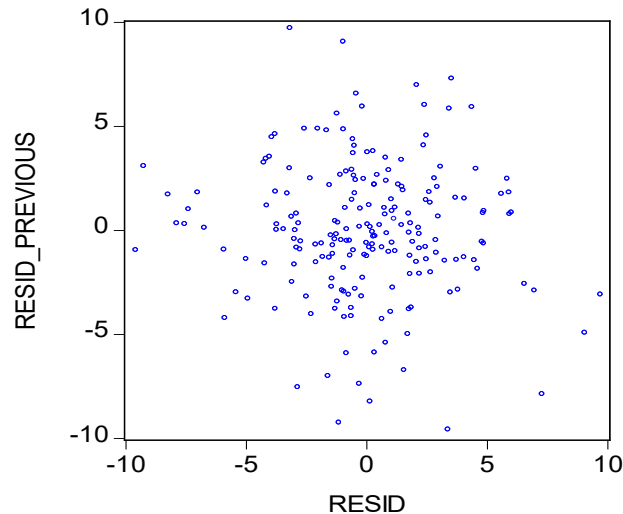
Step 4: Regress using $\hat{u}^2$ and the x's on the right

- If the level of significance is set as 5%, then we do not reject the null. However, if the significance level is 10%, we can reject the null. But, the conclusion may not be reliable here. Why?
- The remedy for heteroskedasticity is simple. We can use the robust standard procedure estimation.
- Engle (1982) proposed a way to “model” the (conditional) variance. The model is known as the autoregressive conditional heteroskedasticity (ARCH) model, a model widely used in many areas of finance, for example Value-at-Risk, option pricing, and _____. [Financial Econometrics]

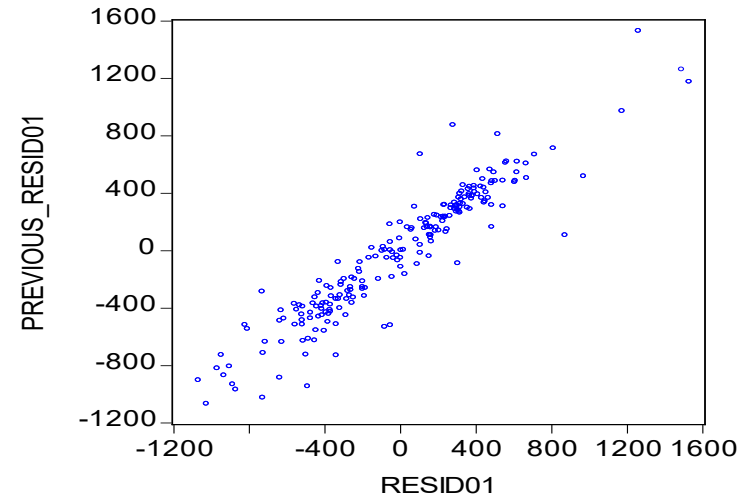
(B) Serial Correlation

- When a time series regression model suffers serial correlation (but TS. Assumptions 1 to 4 are satisfied), the consequences are similar to that of heteroskedasticity.
- OLS estimators are still linear and unbiased, but the standard errors are wrong. Usually, the wrong standard errors tend to be smaller than the correct standard errors, and thus inflate the t values. We tend to over-reject the null hypothesis that the explanatory variable is insignificant.
- If $Cov(u_t, u_{t-j}) = 0$, where $j \neq 0$, then the errors from different time periods are uncorrelated with one another. Graphically, we cannot identify any obvious patterns from the residuals.

Example: Graphical method



(no autocorrelation)



(autocorrelation; serially correlated)

- If we suspect the regression contains first-order autocorrelation with the errors, then the errors can be modeled as follows.

$$u_t = \rho u_{t-1} + e_t, \quad t = 2, \dots, n$$

where $|\rho| < 1$ (remark: it is the stability condition and we will get back to this point), the $E(e_t | u_{t-1}, u_{t-2}, u_{t-3}, \dots) = 0$, and the variance of e_t is σ_e^2 .

- The usual t test is used to test for the following null and alternative hypotheses

$$H_0: \rho = 0 \text{ versus } H_0: \rho \neq 0$$

If the strict exogeneity assumption holds and the sample size is large, we can use $\hat{u}_t$ for u_t

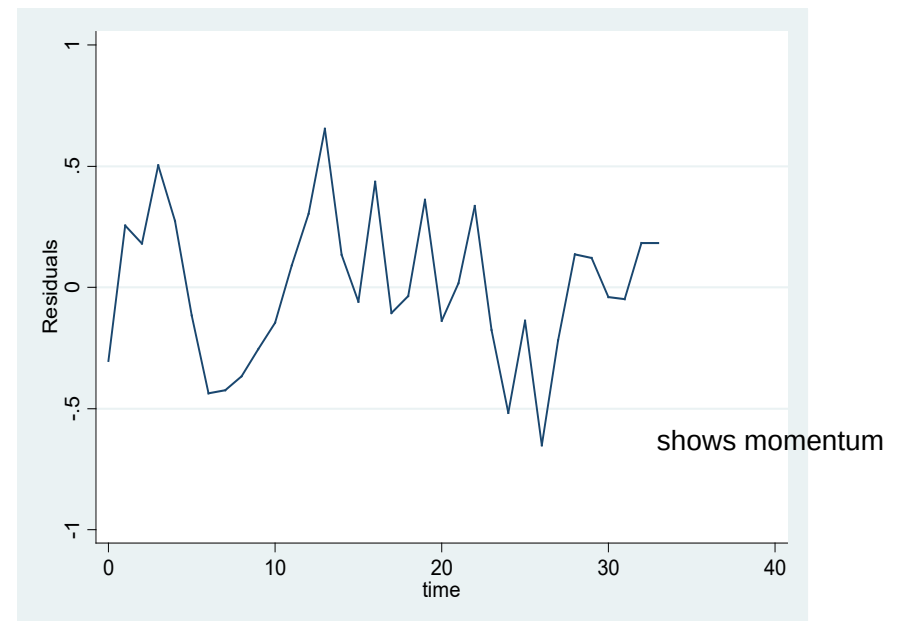
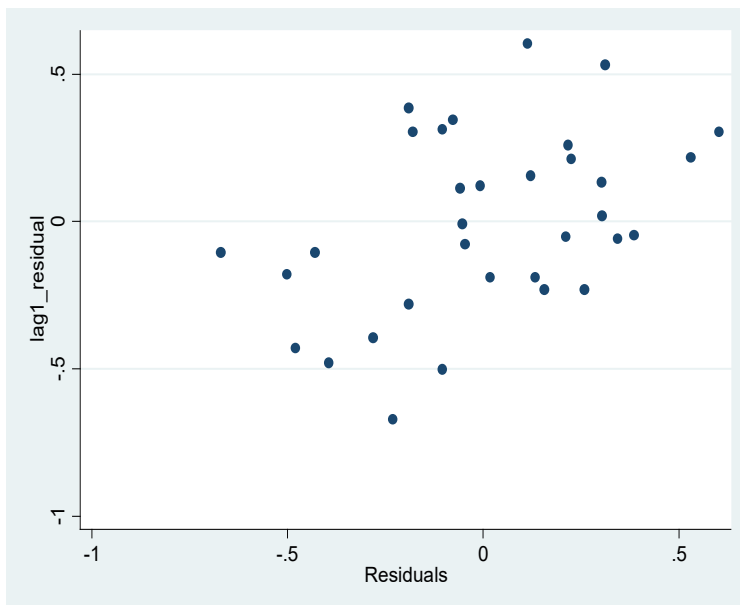
- When testing the null hypothesis, we can use the heteroskedasticity-robust t statistic for the case that e_t is heteroskedastic.
- If we cannot reject the null hypothesis at the 5% significance level, then we can conclude that No Serial Correlation assumption is a reasonable assumption for the model.
- But, if we reject the null hypothesis, then it implies the error terms are serially correlated. Assumption TS 5 does not hold.

Example:

When price of sugarcane is high, farmers plant more than when its price is low. We are therefore interested in below regression model.

$$\ln(Area_t) = \beta_0 + \beta_1 \ln(price_t) + u_t$$

Information on β_1 is useful for government planning. Because, it is important to know whether existing sugar processing mills are likely to be able to handle predicted output, whether a pricing policy linking production, processing, and consumption is desirable.



Question: Do you find any pattern between $\hat{u}_t$ and $\hat{u}_{t-1}$?

The usual t test

Step 1: > **reg** Lland Lprice

Step 2: > **predict** uhat, residual

Step 3: > **gen** uhat_lag_one=ll.uhat

Step 4: > **reg** uhat uhat_lag_one (when sample size is large, reg uhat uhat_lag_one, r)

Source	SS	df	MS	Number of obs = 33		
Model	.477841526	1	.477841526	F(1, 32)	=	6.21
Residual	2.46190253	32	.076934454	Prob > F	=	0.0181
Total	2.93974405	33	.089083153	R-squared	=	0.1625
				Adj R-squared	=	0.1364
				Root MSE	=	.27737

uhat	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
uhat_lag_one	.3992413	.1601969	2.49	0.018	.072931	.7255517

$H_0: \rho = 0$ against $H_0: \rho \neq 0$

$$t_{\hat{\rho}} = \frac{0.399 - 0}{0.16} \approx 2.49$$

We reject the null hypothesis and conclude that serial correlation is a problem to be dealt with at the 5% significance level. Or, since the significance level is bigger than the P value, we reject the null.

The Breusch- Godfrey (BG) Test

- The usual t test (and Durbin–Watson test) is only used to test whether consecutive errors are related to one another, and no lagged dependent variables are allowed in the main regression model. To avoid some of the pitfalls of the usual test of autocorrelation, Breusch and Godfrey proposed a more general test that can be used in a certain set of circumstances.

Example: Even if $Corr(\hat{u}_t, \hat{u}_{t-1}) = 0$, it may be the case that $Corr(\hat{u}_t, \hat{u}_{t-2}) \neq 0$. The *DW* test cannot detect this kind of serial correlation problem.

- The BG test generalizes to any order of autocorrelation, i.e., serial correlation. Additionally, the BG test allows the main regression models to contain lagged dependent variables.

- Consider a regression model:

$$y_t = \beta_0 + \beta_1 x_{t1} + \cdots + \beta_K x_{tK} + u_t, \quad t = 1, \dots, n,$$

Suppose we suspect that error terms are serial correlated with an order p .

$$u_t = \rho_1 u_{t-1} + \cdots + \rho_p u_{t-p} + e_t, \quad e_t \sim N(0, \sigma_e^2)$$

How do we test of the no serial correlation assumption?

- The difficult part is to choose the number of lags, i.e., p . The intuition suggests us to go for a large p . But, for each additional lag, we lose one observation, a large p reduces the number of observations and affects the power of the test.
- In the *BG* test, we test if all $\rho_1, \rho_2, \dots, \rho_p$ are jointly zero.

$H_0: \rho_1 = 0, \rho_2 = 0, \dots, \rho_p = 0$ vs. $H_1: \text{at least one of the } \rho \text{ is not zero}$

$$u_t = \rho_1 u_{t-1} + \cdots + \rho_p u_{t-p} + \sum_{j=1}^k \alpha_j x_{tj} + e_t$$

- We can compute the F test for joint significance of $\hat{u}_{t-1}, \hat{u}_{t-2}, \dots, \hat{u}_{t-p}$. This test may not be able to distinguish which lagged residual is important. In Moreover, this test is only valid asymptotically (i.e., it is valid when we have a large sample size).
- An alternative way to test for the null hypothesis is to use

$$LM = (n - p)R_{\hat{u}_t}^2.$$

Under the null hypothesis $LM \xrightarrow{a} \chi_q^2$.

- If the LM statistic $\geq \chi_{\alpha, q}^2$, we can reject the null.

- Remarks:

(a) If strict exogeneity assumption holds, we can simply omit $\sum_{j=1}^k \alpha_j x_{tj}$, because each explanatory variable x_{tj} is not correlated to the error terms $u_{t-1}, u_{t-2}, \dots, u_{t-p}$

(b) If the main regression model contains lagged dependent variables, we should include all explanatory variables and run the following regression model and test

$H_0: \rho_1 = 0, \rho_2 = 0, \dots, \rho_p = 0$, against $H_1: H_0$ is wrong.

$$u_t = \rho_1 u_{t-1} + \dots + \rho_p u_{t-p} + \sum_{j=1}^k \alpha_j x_{tj} + e_t$$

Example: Consider the model $\ln(\text{Area}_t) = \beta_0 + \beta_1 \ln(\text{price}_t) + u_t$. Suppose we suspect u_t is serially correlated with its past 2 periods, u_{t-1} and u_{t-2}

Step 1: > **reg** Lland Lprice

Step 2: > **predict** uhat, **residual**

Step 3: > **gen** uhat_lag_one= l1.uhat; gen uhat_lag_two= l2.hat;

Step 4: > **reg** uhat uhat_lag_one gen uhat_lag_two, **noconstant**

. reg uhat l1.uhat l2.uhat, noconstant

Source	SS	df	MS	Number of obs = 32		
Model	.56804099	2	.284020495	F(2, 30) = 3.69		
Residual	2.30696512	30	.076898837	Prob > F = 0.0368		
Total	2.87500611	32	.089843941	R-squared = 0.1976		
				Adj R-squared = 0.1441		
				Root MSE = .27731		

uhat	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
uhat						
L1.	.4631294	.1771813	2.61	0.014	.1012768	.824982
L2.	-.0618053	.1754274	-0.35	0.727	-.4200758	.2964653

At 10% significance level, we reject $H_0: \rho_1 = 0, \rho_2 = 0$. Because, the P value for the F statistic is 0.0368.

We use the LM test. $LM = (n - p)R_{\hat{u}_t}^2 = (32 - 2) \times 0.197 = 5.91 > \chi_{10\%}^2(2) = 4.61$.

Reject the null as well.

(C) Correcting for Serial Correlation

- If Assumption TS. 5: No Serial Correlation is violated, how to fix this autocorrelation problem?
 - (a) Use Feasible Generalized Least Squares (FGLS) instead of OLS.
 - (b) The autocorrelation is often caused by a dynamic structure in the dependent variable that has not been modeled and captured in the model. Therefore, we can include lagged dependent variables Y or even X
- To use the FGLS method, we need to have strictly exogenous explanatory variables. Therefore, at a minimum, we should not use the following methods when the explanatory variables include lagged dependent variables.

$$\text{Var}(y_t) = \text{Var}(\rho y_{t-1} + e_t) = \rho^2 \text{Var}(y_{t-1}) + \text{Var}(e_t) + 2\rho \text{Cov}(y_{t-1}, e_t)$$

- Obtaining the Linear Unbiased Estimator in the AR(1) Model

For the sake of simplicity, we consider a simple regression model

$$y_t = \beta_0 + \beta_1 x_t + u_t,$$

and suppose the error is first-order autocorrelated, then

$$u_t = \rho u_{t-1} + e_t, \quad t = 1, 2, \dots, n$$

$$\text{Var}(y_t) = \rho^2 \text{Var}(y_t) + \sigma_e^2$$

$$(1 - \rho^2) \text{Var}(y_t) = \sigma_e^2$$

where serially uncorrelated $e_t \sim N(0, \sigma_e^2)$ and $|\rho| \leq 1$. Additionally, we assume the Gauss-Markov assumptions TS.1 through TS.4 are satisfied. Note that u_t is

a stationary series and its $\text{Var}(u_t) = \frac{\sigma_e^2}{1 - \rho^2}$

$$\text{Var}(u_t) = \frac{\sigma_e^2}{1 - \rho^2}$$

For $t \geq 2$, we can write the models as

$$(y_t - \rho y_{t-1}) = (1 - \rho)\beta_0 + \beta_1(x_t - \rho x_{t-1}) + (u_t - \rho u_{t-1})$$

$$y_t^* = \beta_0^* + \beta_1 x_t^* + e_t, \quad t = 2, \dots, n \text{-----} (*)$$

where $y_t^* = y_t - \rho y_{t-1}$, $x_t^* = x_t - \rho x_{t-1}$, and $\beta_0^* = (1 - \rho)\beta_0$

- If we omit the first observation ($t = 1$), the OLS estimators are not BLUE. We could consider adding the first observation when we are estimating the model.
- For $t = 1$ case, based on (*) we are not able to compute y_t^* and x_t^* as we don't have y_{t-1} and x_{t-1} , because $y_1 = \beta_0 + \beta_1 x_1 + u_1$.

Since $Var(u_1) = \frac{\sigma_e^2}{1-\rho^2}$, We can transform the model to

$$\begin{aligned} \sqrt{(1-\rho^2)}y_1 &= \sqrt{(1-\rho^2)}\beta_0 + \beta_1\sqrt{(1-\rho^2)}x_1 + \sqrt{(1-\rho^2)}u_1 \\ \Rightarrow \tilde{y}_1 &= \beta_0\tilde{x}_0 + \beta_1\tilde{x}_1 + \tilde{u}_1, \text{ where } \tilde{y}_1 = \sqrt{(1-\rho^2)}y_1 \text{ and } \tilde{x}_1 = \sqrt{(1-\rho^2)}x_1 \end{aligned}$$

- In summary, we need to transform the original y and X data so that the “new” error terms in the model are serially uncorrelated.

$$\text{For } t = 1, \tilde{y}_1 = \sqrt{(1-\rho^2)}y_1 \text{ and } \tilde{x}_1 = \sqrt{(1-\rho^2)}x_1$$

$$\text{For } t \geq 2, y_t^* = y_t - \rho y_{t-1} \text{ and } x_t^* = x_t - \rho x_{t-1}$$

Once we have the new y and X data, we can run the OLS regression model as usual.

- But there is another practical problem here: we don't know the value of ρ .
- Feasible generalized least squares estimation with AR(1) errors: We use $\hat{\rho}$ to substitute ρ and then run the regression model based on “new” data $y_t^*, x_t^*, \tilde{y}_1, \tilde{x}_1$. The estimators are called FGLS estimators of β_j .
- There are several names for FGLS estimation of the AR(1) model that come from different methods of estimating ρ and different treatment of the first observation.
 - (i) Cochrane-Orcutt (CO) estimation omits the first observation and uses $\hat{\rho}$ estimate from $u_t = \rho u_{t-1} + e_t$
 - (ii) Prais-Winsten (PW) estimation uses the first observation as previously suggested way, and obtains $\hat{\rho}$ estimate from $u_t = \rho u_{t-1} + e_t$
- Asymptotically, both the CO and PW estimations method are indifferent. However, in practice since many time series samples are small, the differences can be notable in applications.

Example: Consider the model $\ln(Area_t) = \beta_0 + \beta_1 \ln(price_t) + u_t$. We use the CO estimation.

Step 1: Run the original regression model `> reg y x`

Step 2: Calculate the residual `> predict uhat, residual`

Step 3: Create a lagged residual `> gen uhat_1=l1.uhat`

Step 4: Use the $\hat{\rho}$ to transform the y and x data (the $\hat{\rho}$ in this example is 0.3999)

➤ `gen rhoY= $\hat{\rho} \times l1.y$` (note that it is ρy_{t-1})

➤ `gen y*=y - rhoY`

➤ `gen rhoX= $\hat{\rho} \times l1.x$` (note that it is ρx_{t-1})

➤ `gen x*=x - rhoX`

Step 5: run y^* on x^* `> reg y* x*`

`. reg y_star x_star`

Source	SS	df	MS	Number of obs = 33		
Model	.934941417	1	.934941417	F(1, 31) = 11.85		
Residual	2.44493847	31	.078868983	Prob > F = 0.0017		
Total	3.37987988	32	.105621246	R-squared = 0.2766		
				Adj R-squared = 0.2533		
				Root MSE = .28084		

y_star	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
x_star	.8783761	.2551182	3.44	0.002	.3580591	1.398693
_cons	2.339877	.0526856	44.41	0.000	2.232424	2.44733

This estimated intercept 2.339 is for β_0^* .

We can convert it back for β_0 . The estimate will be 3.897.

- In Stata, we can use the command “corc” if we want to use the CO estimation method.

> CORC Lland Lprice

```
. corc Lland Lprice
Iteration 0: rho = 0.0000
Iteration 1: rho = 0.3992
Iteration 2: rho = 0.4201
Iteration 3: rho = 0.4220
```

(Cochrane-Orcutt regression)

Source	SS	df	MS			
Model	.971240631	1	.971240631	Number of obs =	33	
Residual	2.44357498	31	.078824999	F(1, 31) =	12.32	
Total	3.41481561	32	.106712988	Prob > F =	0.0014	
				R-squared =	0.2844	
				Adj R-squared =	0.2613	
				Root MSE =	.28076	

Lland	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
Lprice	.8882951	.2530614	3.51	0.001	.372826	1.403764
_inter	3.898774	.0906011	43.03	0.000	3.714225	4.083322
_cons	(omitted)					
rho	0.4221	0.1590	2.66	0.012	0.0983	0.7459

```
Durbin-Watson statistic (original) 1.168987
Durbin-Watson statistic (transformed) 1.820534
```

The estimated coefficient for the explanatory is 0.888 which is very close to 0.878 (the estimate we got from step by step). Note that the estimated intercept here is for the original intercept.

- In Stata, we can use the command “prais” if we want to use the PW estimation method.

>**prais** Lland Lprice

Prais-Winsten AR(1) regression -- iterated estimates

Source	SS	df	MS	Number of obs = 34		
Model	1.49591777	1	1.49591777	F(1, 32) = 19.18		
Residual	2.49534794	32	.077979623	Prob > F = 0.0001		
Total	3.99126571	33	.120947446	R-squared = 0.3748		
				Adj R-squared = 0.3553		
				Root MSE = .27925		

Lland	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
Lprice	.9561701	.2383577	4.01	0.000	.4706514	1.441689
_cons	3.872848	.0859255	45.07	0.000	3.697824	4.047873
rho	.4329136					

Durbin-Watson statistic (original) 1.168987
Durbin-Watson statistic (transformed) 1.920066

The estimated coefficient for explanatory variable is 0.956. This estimate is a bit different from the estimate from the CO estimation. Why so?

- Remark:

The practical problem facing the researcher is this: In the presence of autocorrelation, OLS estimators, although unbiased, consistent, and asymptotically normally distributed, are not efficient. Therefore, the usual inference procedure based on the t , F , and χ^2 tests is no longer appropriate. On the other hand, FGLS produces estimators that are efficient, but the finite, or small-sample, properties of these estimators are not well documented. This means in small samples the FGLS might actually do worse than OLS. As a matter of fact, in a Monte Carlo study Griliches and Rao found that if the sample is relatively small and the coefficient of autocorrelation, ρ , is less than 0.3, OLS is as good or better than FGLS. As a practical matter, then, one may use OLS in small samples in which the estimated ρ is, say, less than 0.3. Of course, what is a large and what is a small sample are relative questions, and one has to use some practical judgment. If you have only 15 to 20 observations, the sample may be small, but if you have, say, 50 or more observations, the sample may be reasonably large. (Gujarati, 2008)